

GateFlow

Smart traffic congestion alert for campus arrivals, 08:00–10:00

Forecasts Low / Moderate / High one hour ahead · explains why · alerts before the jam

VIT AI Innovate-A-Thon · Question 2 · October 2026
Proof of concept on real public traffic data · no live traffic feed connected

Students all arrive 08:00–10:00 and the gate road jams

2 hours

when most of campus arrives at the same gate

23%

of weekday 8–10 AM hours reach High congestion in our data

Congestion depends on vehicle volume, weather and events. A warning only helps if it comes before the jam.



Transport / admin office

Decides on extra shuttles, a second gate and marshals. Gets the 07:00 alert with actions.



Gate security staff

Needs to know when crowds peak. Gets a simple level plus the reason.



Students and staff

Want to arrive on time. Get a "leave 20 minutes earlier" message via the office.



Shuttle operators

Plan trips around peaks. Get forecasts for every weekday, as CSV.

A one-hour-ahead forecast that explains itself



Forecast, not hindsight

At 07:00 it predicts 08:00–09:00, using only what is known at 07:00. It never sees the answer.



Explains every forecast

Shows how much recent traffic, usual traffic, weather and holidays push the chance of High up or down.



AI that runs in the browser

The trained trees are exported to JSON and run in JavaScript, identical to scikit-learn. Free hosting, instant, private.



Built for real use

Replay any day, what-if sliders, type today's numbers, upload a CSV and download forecasts.

Real data: 6 years of hourly traffic, weather and holidays

Task field	Dataset column
Timestamp	date_time (hourly, Oct 2012 – Sep 2018)
Vehicle count	traffic_volume (vehicles per hour)
Weather	weather_main + temp, rain, snow, clouds
Event indicator	holiday (US holidays, State Fair)

UCI Machine Learning Repository / Kaggle · 48,204 rows · CC BY 4.0 · highway I-94 used as a stand-in for the campus road

Labels (volume-to-capacity proxy)

Low	< 5,000 veh/h
Moderate	5,000 – 5,800
High	≥ 5,800 (≈85% of capacity)

Preprocessing

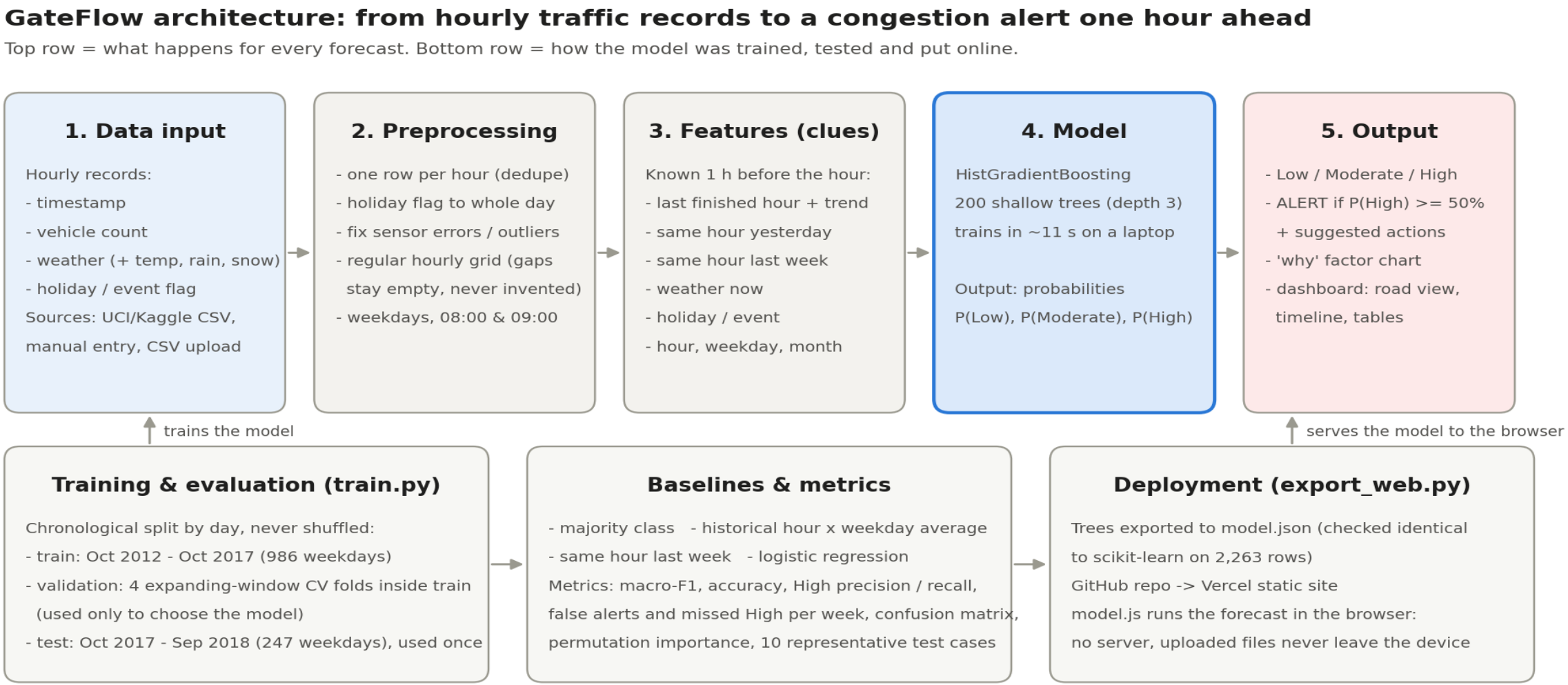
- Dedupe hours (7,629 duplicate rows), spread the holiday flag to the whole day
- Fix sensor errors; the 2014–15 outage stays empty, never invented
- Weekdays only, target hours 08:00 and 09:00

Chronological split, never shuffled



Validation = 4 expanding-window CV folds inside train, used only to choose the model. Test = last year, 247 days, 494 forecasts, used once.

Architecture: data in, forecast and alert out



A lightweight model: 200 small trees, trained in 11 seconds

Clues known one hour before the arrival hour

- Traffic in the last finished hour, and its trend
- Same hour yesterday and last week
- Weather now: type, temperature, rain, snow, cloud
- Holiday or event that day
- Arrival hour, weekday, month

Example on 12 Sep 2018: 5,908 cars at 6–7 AM, up 2,823 on the hour before, rain → 99% chance of High.

HistGradientBoosting

200 trees, 3 questions deep, each fixing the last one's mistakes

11 s

training time on a laptop CPU: no GPU, no deep learning

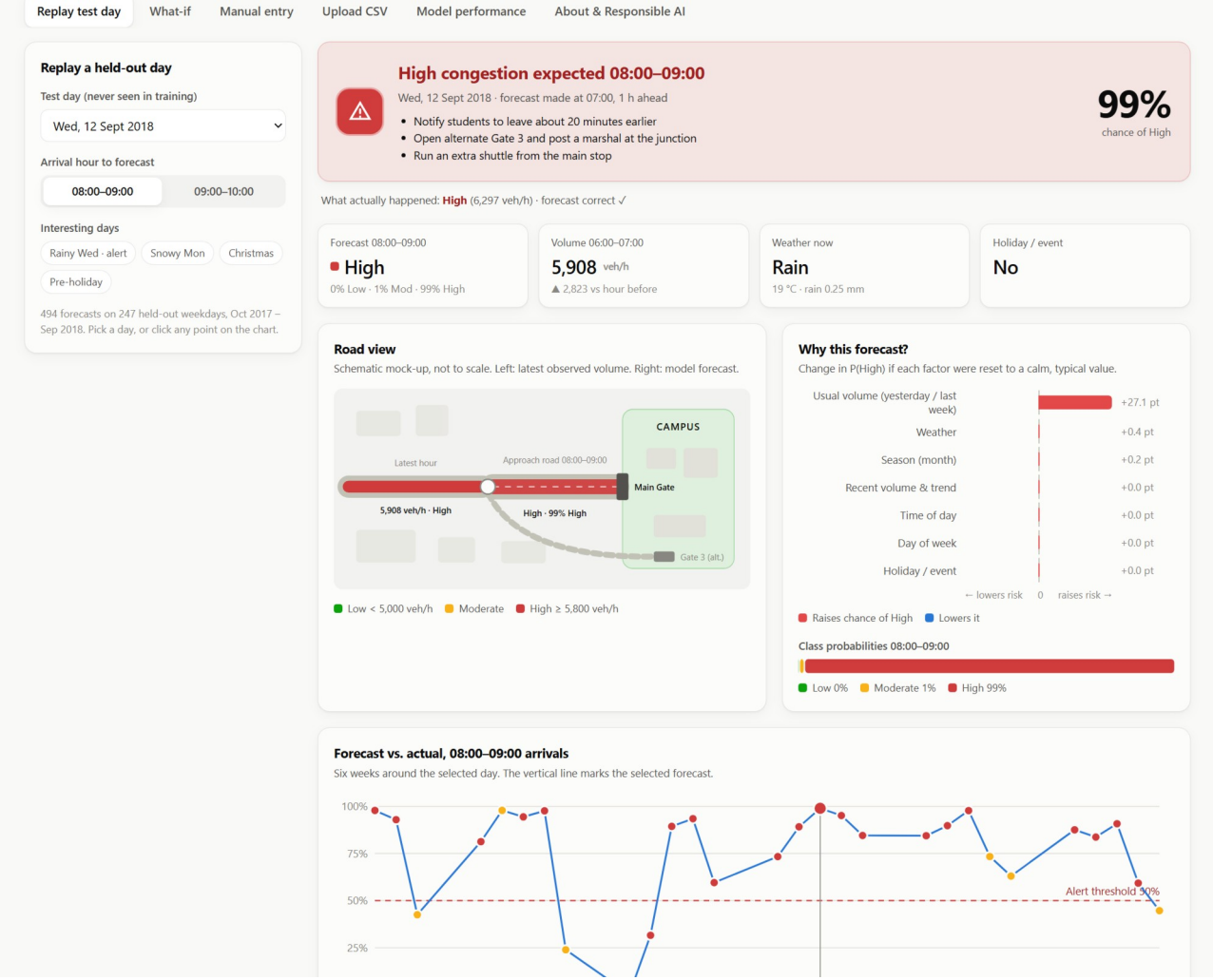
0.705 vs 0.672

validation macro-F1 vs logistic regression, so it was picked before testing

3×10^{-7}

largest gap between browser and scikit-learn outputs

Demo: a day the model never saw, caught one hour early



1

Pick a held-out day

Wed 12 Sep 2018, rainy. Forecast made at 07:00.

2

Red alert + actions

99% chance of High: leave earlier, open Gate 3, extra shuttle.

3

See why

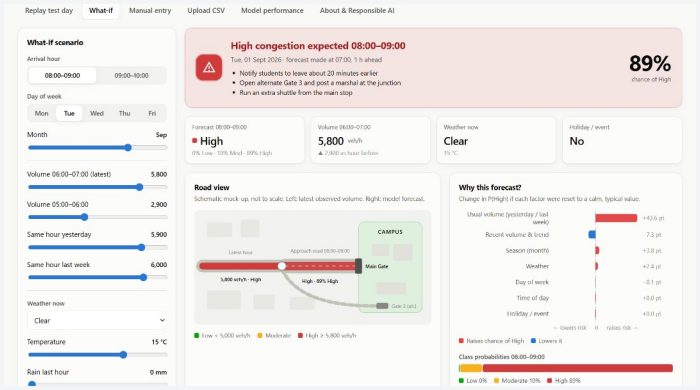
Usual traffic for this slot adds +27 points to the chance of High.

4

Check the truth

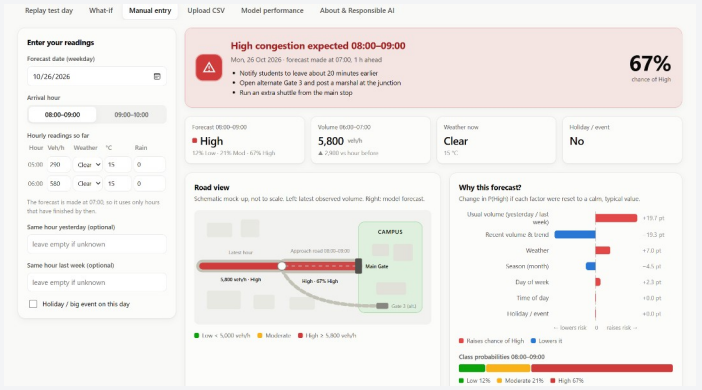
Actual: 6,297 veh/h, High. The forecast was correct.

User workflow: three ways to get a forecast



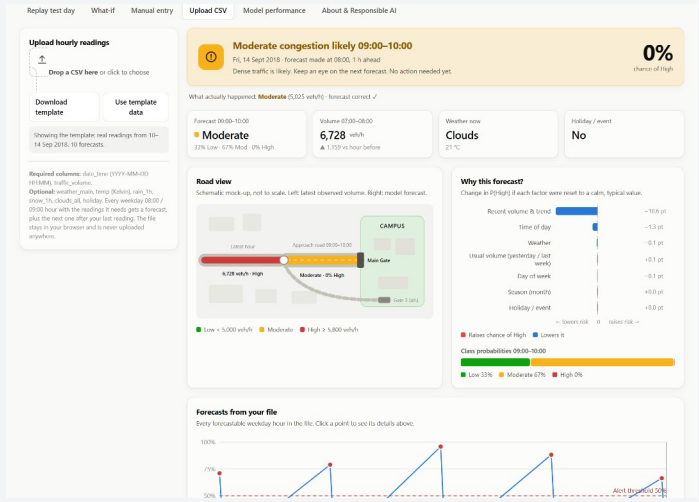
What-if

Drag sliders for traffic, weather, month or holiday. The alert updates live.



Manual entry

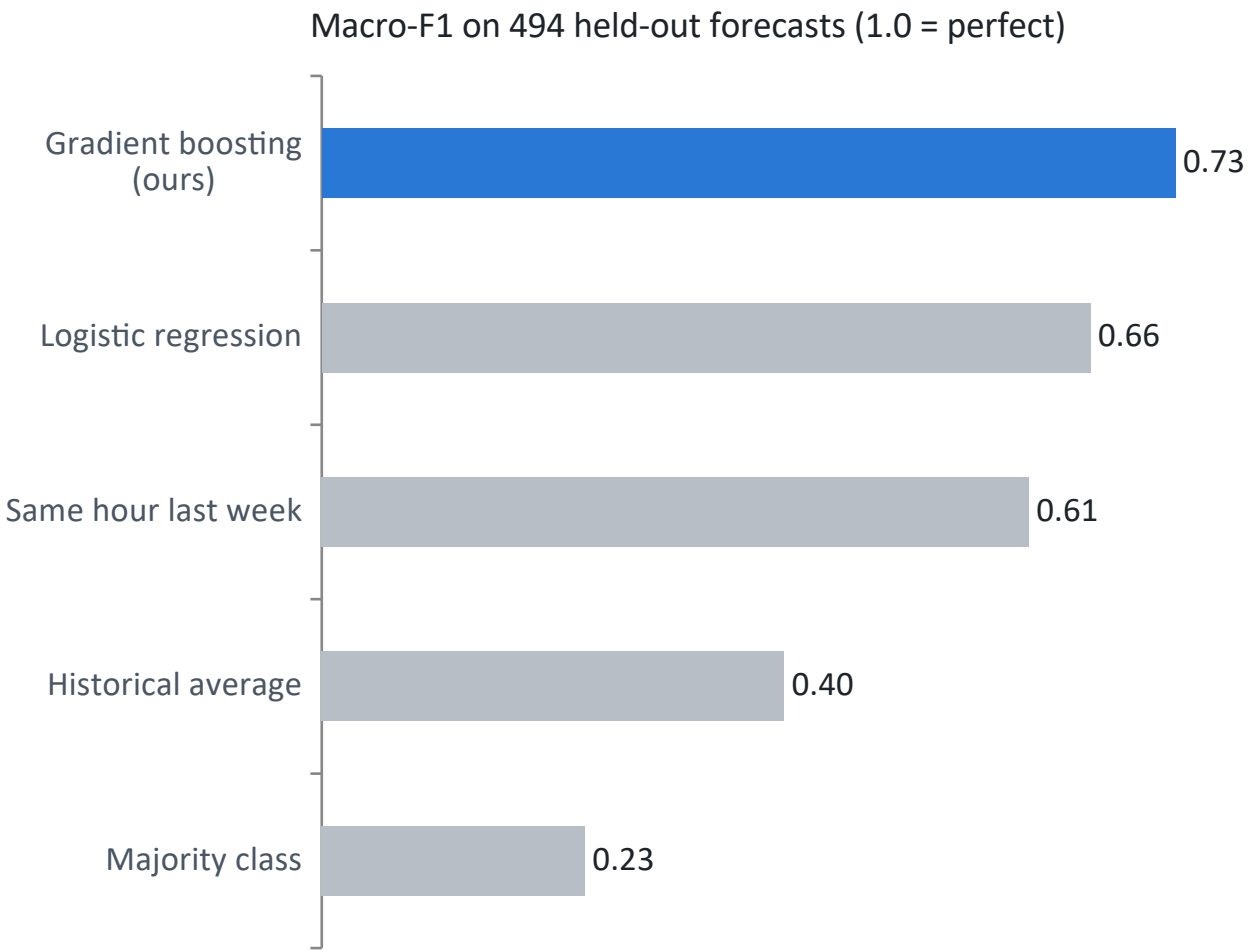
Type today's hourly counts and weather into a table, then get the 08:00 or 09:00 forecast.



Upload CSV

Drop a file: every weekday peak in it gets a forecast, with a table and CSV download.

On a year it never saw, it beats every simple rule



0.73

macro-F1, vs 0.61 for the best simple rule

80%

of High-congestion hours caught

~1

false alert per week (out of 10 forecasts)

0.69

macro-F1 even 2 hours ahead (92% of High caught)

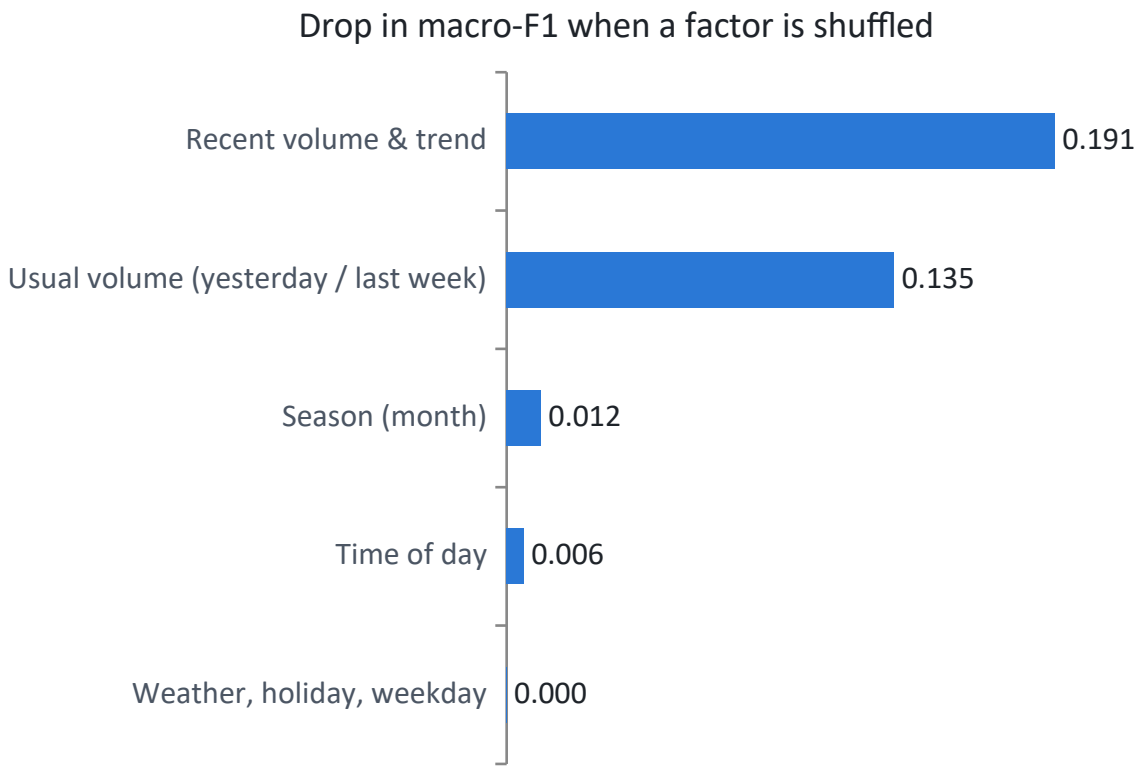
Test cases: where it works and where it fails

Arrival hour	Result	Situation	P(High)	Predicted → actual
12 Sep 2018 08:00	**Success**	Rainy peak: alert raised and correct	99%	High → High (6,297)
23 Aug 2018 08:00	**Success**	Event day (State Fair), heavy traffic	93%	High → High (6,162)
25 Dec 2017 08:00	**Success**	Christmas: no false alarm	0%	Low → Low (1,260)
05 Feb 2018 08:00	**Failure**	Quiet 6 AM, then a sudden surge	1%	Low → High (6,048)
19 Apr 2018 09:00	**Failure**	Barely over the High cut-off	0%	Moderate → High (5,842)
28 Aug 2018 08:00	**Failure**	False alarm: ended just under the cut-off	98%	High → Moderate (5,692)

Error analysis

- 79% of false alarms ended within 300 veh/h below the cut-off; all 20 missed High hours were within 300 above it
- Weakest in fog and rain (67–68% correct), best in snow (91%); all 20 holiday forecasts were right

What drives the forecast, and what it can't do yet



Weather and holidays matter on the days they occur, but they are rare in the test year. The per-forecast "why" chart shows their effect.

Limitations

Not live

Replays history; no traffic or weather API yet

Stand-in road

Highway I-94, not our gate: retrain on our counts

Volume, not speed

Labels are a proxy; queues and accidents are invisible

Hourly

15-minute data would give earlier, sharper alerts

No campus events

Exams and fests are not in the data

Responsible AI: private, fair about its limits, human in charge

Privacy

Vehicle counts only. No cameras, plates or personal data. Uploaded CSVs never leave the browser.

Bias & fairness

One US highway; weaker in fog and rain (67–68%). Balanced class weights so High isn't ignored.

Security

Static site: no server, database or keys. User text is escaped; files are parsed as plain data.

Accessibility

Word + icon for every level, colour-blind-safe colours, keyboard use, screen-reader labels, dark mode, mobile.

Limitations

Proxy labels, stand-in road, hourly data, no live feed. All stated on the site and in the report.

Human oversight

Advisory only: the transport office decides. Borderline 40–60% shown as a probability. Nothing automated.

Aligned with the UN Sustainable Development Goals



SDG 11: Sustainable cities & communities

Target 11.2: safer, more accessible transport at a shared choke point; pushes shuttles and alternate gates.



SDG 3: Good health & well-being

3.6 road injuries and 3.9 air pollution: calmer gate queues, less stop-start exhaust.



SDG 13: Climate action

Smoother flow and more shuttle use mean less idling and lower emissions.



SDG 9: Industry, innovation & infrastructure

9.1: more out of existing roads through data, not new construction.



SDG 4: Quality education

4.a: students reach class on time and safely.

Implemented today, and what comes next

Implemented and working

- Real-data model, beats all baselines on a held-out year
- 1-hour-ahead forecast, alert at P(High) ≥ 50%, explanations
- Web dashboard: replay, what-if, manual entry, CSV upload, performance, responsible AI
- Model runs in the browser; deployed from GitHub to Vercel
- Metrics, confusion matrix, test cases with failures, parity tests

Planned next

- Count vehicles at our own gate for a few weeks and retrain (same code)
- Speed or queue labels from a camera or GPS
- Live traffic counter and weather-forecast APIs
- SMS or app notifications run by the transport office
- 15-minute data and the campus event calendar



Catches 4 of 5 jams, one hour early

About one false alarm a week · explainable · private · runs free in any browser

Code: github.com/AnmolDiwan/gateflow-congestion-alert
Data: Hogue (2019), Metro Interstate Traffic Volume, UCI ML Repository, CC BY 4.0